Status	Finished
Started	Thursday, 20 November 2025, 1:31 PM
Completed	Thursday, 20 November 2025, 1:46 PM
Duration	15 mins 21 secs
Marks	3.00/3.00
Grade	**3.00** out of 3.00 (**100**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Correct

Mark 1.00 out of 1.00

If p and q be prime numbers, let $n = p \cdot q$, and write (n, x) and (n, y) for the public and private key in an RSA key pair associated to n.

In the list below we have written suggestions for $(x, y, p, q, p \cdot q)$. Some of them give valid RSA keys, some of them do not for different reasons.

Select all options that give us a valid key pair. Leave the invalid options un-checked.

- [ ] **x** = 5066098, **y**=11977289, **p**=6709, **q**=5641, **n**=37845469
- [x] **x** = 682183, **y**=22904887, **p**=6163, **q**=4217, **n**=25989371
- [ ] **x** = 3008046, **y**=2067287, **p**=3881, **q**=2549, **n**=9892669
- [ ] **x** = 26924209, **y**=43485577, **p**=7239, **q**=7789, **n**=56384571
- [x] **x** = 36661763, **y**=31000091, **p**=5783, **q**=6553, **n**=37895999
- [ ] **x** = 7637383, **y**=15705847, **p**=3921, **q**=5531, **n**=21687051
- [ ] None of the above

You are free to use this calculator for products $\mod (n)$. For example, if you want to compute $(7634 \cdot 99833\%8883993$, you should set **Factor 1**=7634, **Factor 2**=99382, and **Modulus**=8883993.

```
Factor 1:
factor 2:
Modulus:
Result:
```

You are free to use this calculator for computing powers $\mod (n)$. For example, if you want to compute $(7634^{99382})\%8883993$, you should set **Base**=7634, **Exponent**=99382, and **Modulus**=8883993.

```
Base:
Exponent:
Modulus:
Result:
```

Question **2**

Correct

Mark 1.00 out of 1.00

Compute the minimal and maximal Hamming distance between any two bitstrings in the set

$$[000111, 000011, 100101, 010011]\,.$$

$l_{min} =$ 1

Your last answer was interpreted as follows:

$$1$$

$l_{max} =$ 4

Your last answer was interpreted as follows:

$$4$$

Question **3**

Correct

Mark 1.00 out of 1.00

Let

$$E : (\mathbb{Z}/2)^{\times 3} \to (\mathbb{Z}/2)^{\times 5}$$

be the encoding function given by $E(w_1 w_2 w_3) = w_1 w_2 w_3 w_4 w_5$, where

$$\begin{bmatrix} w_4 = w_3 + w_2 + w_1 \\ w_5 = 0 \end{bmatrix}$$

a) Find the generator matrix G for the encoding E

$G =$

```
1 0 0 1 0
0 1 0 1 0
```

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

b) Find the parity check matrix H for the encoding E.

$H =$

```
1 1 1 1 0
0 0 0 0 1
```

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

c) Let $r = 00010$ be a received message. Compute the syndrome of r with respect to the encoding E.

Syndrome of r = 1 0

Your last answer was interpreted as follows:

$$1\,0$$

Syntax guide: Give your answers to **a)** and **b)** as matrices. Use a space character to separate entries in the same row, and line break to separate rows.

Syntax guide: Give your answer to **c)** as a bitstring. E.g. if you think the answer to a question is "1011", then you should enter **1011** .

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 522

Signing code: 48624

Your last answer was interpreted as follows:

$$48624$$

◄ Test 3 (topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod(n) and Lagrange's theorem)

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